

PURE MATHEMATICS WITH MECHANICS • UPPER SIXTH

UPPER SIXTH — PURE MATHEMATICS WITH MECHANICS

CHAPTER 3

VECTORS

SUGGESTED TIME: 11 PERIODS

IN THIS CHAPTER

This chapter extends vector methods to the geometry of planes in three-dimensional space. Learners will derive vector, parametric and Cartesian equations of planes, compute distances and angles, and study the intersection of lines and planes — essential tools for the GCE Advanced Level examination.

- By the end of this chapter I can write the vector equation of a plane given a point and a normal vector, or a point and two direction vectors.
- By the end of this chapter I can convert between the vector, parametric and Cartesian forms of the equation of a plane.
- By the end of this chapter I can find the Cartesian equation of a plane through three non-collinear points.
- By the end of this chapter I can calculate the perpendicular distance from a point to a plane.
- By the end of this chapter I can find the angle between two planes and the angle between a line and a plane.
- By the end of this chapter I can determine the point of intersection of a line and a plane and discuss existence and uniqueness.



DOWNLOAD ON- BUCH

Free • Google Play
AI tutor, past papers,
results



GO ONBUCH+

All signed courses, unlimited AI
tutor, PDF sheets — OnBuch+
tab in the app

THIS SIGNED COURSE CONTAINS

COURSE

complete, illustrated,
step by step

METHODS

and worked examples

EXERCISES

exam style + solutions

SUMMARY SHEET

the essentials to
remember



Signed • OnBuch+

ONBUCH+ TEACHING TEAM
FOLLOWS THE OFFICIAL MINESEC SYLLABUS

OnBuch+

CONTENTS

- | | |
|---|---|
| 1. Equations of a Plane: Vector, Parametric and Cartesian Forms | of Two Planes |
| 2. Plane Through Three Points and Special Planes | 6. Pencils of Planes and Synthesis Problems |
| 3. Perpendicular Distance from a Point to a Plane | 7. Integration activity |
| 4. Angles: Between Two Planes and Between a Line and a Plane | 8. Methods and worked examples |
| 5. Intersection of a Line with a Plane and | 9. Exercises |
| | 10. Solutions to the exercises |
| | 11. Summary sheet |

CHAPTER OBJECTIVES

- By the end of this chapter I can write the vector equation of a plane given a point and a normal vector, or a point and two direction vectors.
- By the end of this chapter I can convert between the vector, parametric and Cartesian forms of the equation of a plane.
- By the end of this chapter I can find the Cartesian equation of a plane through three non-collinear points.
- By the end of this chapter I can calculate the perpendicular distance from a point to a plane.
- By the end of this chapter I can find the angle between two planes and the angle between a line and a plane.
- By the end of this chapter I can determine the point of intersection of a line and a plane and discuss existence and uniqueness.
- By the end of this chapter I can find the line of intersection of two planes and recognise parallel or coincident planes.
- By the end of this chapter I can write the equation of a plane through the line of intersection of two given planes (pencil of planes).
- By the end of this chapter I can solve GCE-style problems combining several of these skills.

PREREQUISITES

- Scalar (dot) product of two vectors and its link with the angle between them (Lower Sixth).
- Vector and parametric equations of a straight line in 3D (Lower Sixth).
- Vector (cross) product and its geometric meaning as area and perpendicular direction.
- Position vectors, magnitude of a vector, and basic vector algebra.
- Solving simultaneous linear equations in two and three unknowns.